

Modeling and Control of Cable-Driven UAV Systems for Collaborative Load Manipulation

Chiara Buono

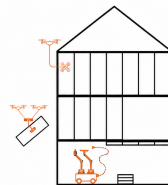
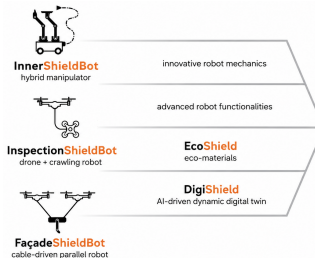
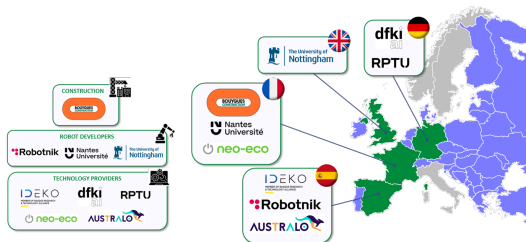
Advisors: Fantoni, Yigit, Simetti

26th August 2026





Image source: AI generated



SUSTAINABLE AND EFFICIENT
CONSTRUCTION, RENOVATION
& MAINTENANCE

Image source: ShieldBot Project Consortium
(2026)

- **State of the Art**
- Kinemato-static Modeling
- Design Space
- Architecture Selection Methodology
 - Feasibility Filters
 - Geometric Performance Indices
 - Simulation Environment
- Control Architecture
- Baseline & Experimental Setup
 - Tension Allocation Strategy
 - Stewart Baseline
 - Statistical Methodology
- Experimental Results and Discussion
 - Static vs. Dynamic
 - Ground Anchor Spacing
 - UAV Attachment Geometry
- Conclusions and Future Work

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Feature	CDPRs
Infrastructure Independence	X
Mechanical Simplicity	✓
Wrench-generation capability	✓

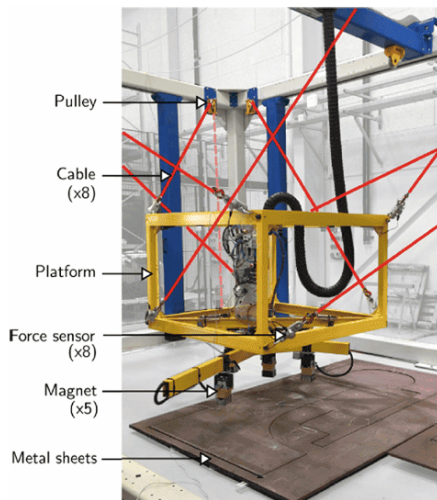


Image source: Pott and Bruckmann (2013), Picard et al. (2018)

Erskine, Chriette, and Caro (2019), Erskine, Chriette, and Caro (2019), Masone, Bülthoff, and Stegagno (2016)

Feature	CDPRs
Infrastructure Independence	✗
Mechanical Simplicity	✓
Wrench-generation capability	✓

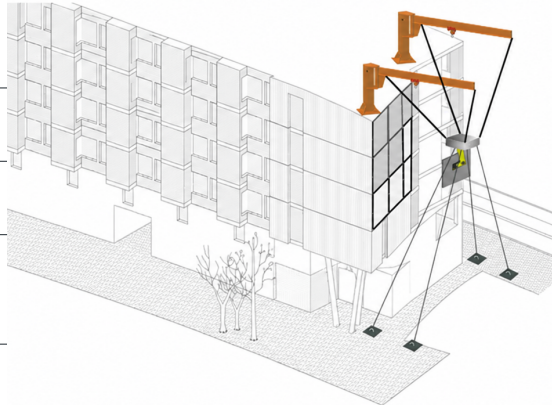


Image source: ShieldBot Project Consortium (2026)
 Erskine, Chriette, and Caro (2019), Masone, Bühlhoff,
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Feature	FPRs
Infrastructure Independence	✓
Mechanical Simplicity	✗
Wrench-generation capability	≈

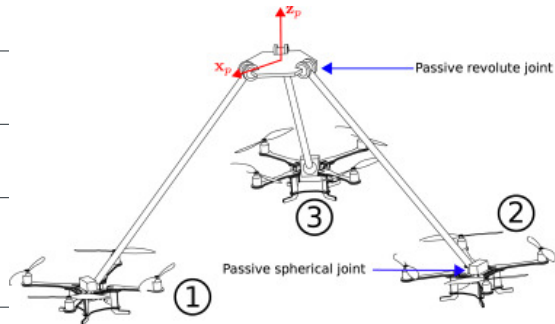


Image source: Six et al. (2021)

Six et al. (2021), Liu et al. (2022)

Feature	Standard ACTS
Infrastructure Independence	✓
Mechanical Simplicity	✓
Wrench-generation capability	✗

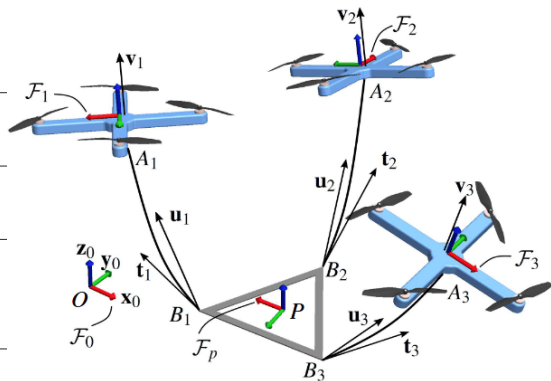


Image source:

Jamshidifar and Khajepour (2020),

Xiong et al. (2022)

Erskine, Chriette, and Caro (2019)

Feature	Hybrid ACTS
Infrastructure Independence	✓
Mechanical Simplicity	✓
Wrench-generation capability	✓

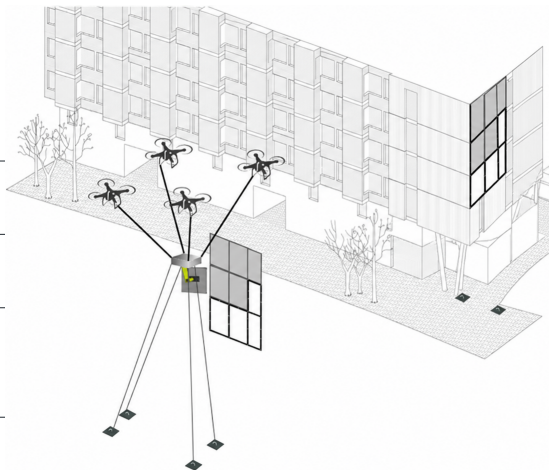
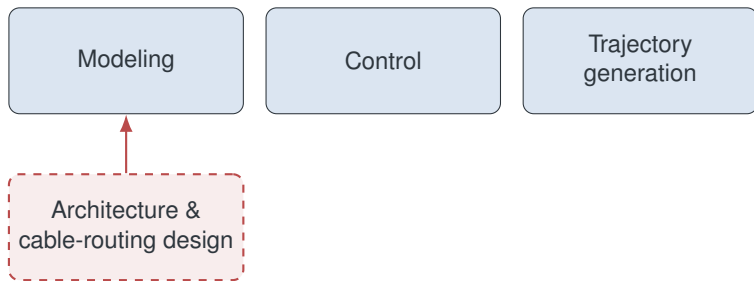


Image source: Jamshidifar and Khajepour (2020), Xiong et al. (2022) ShieldBot Project Consortium (2026)

System Type	Infrastructure Independence	Mechanical Simplicity	Wrench-generation capability
CDPRs	$\times$	$\checkmark$	$\checkmark$
FPRs	$\checkmark$	$\times$	$\approx$
Standard ACTSs	$\checkmark$	$\checkmark$	$\times$
Hybrid ACTSs	$\checkmark$	$\checkmark$	$\checkmark$

Goal: bridge design and control with a reproducible **architecture selection methodology**



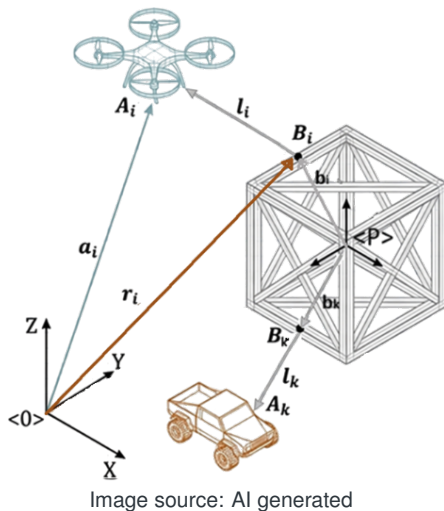
Cable and Anchor Jacobians

$$\mathbf{J}_a = \begin{bmatrix} \mathbf{u}_1^\top & \mathbf{0}^\top & \cdots & \mathbf{0}^\top \\ \mathbf{0}^\top & \mathbf{u}_2^\top & \cdots & \mathbf{0}^\top \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}^\top & \mathbf{0}^\top & \cdots & \mathbf{u}_m^\top \end{bmatrix} \quad (1)$$

$$\mathbf{J}_p = \begin{bmatrix} \mathbf{u}_1^\top & ({}^0\mathbf{R}_P {}^P\mathbf{b}_1 \times \mathbf{u}_1)^\top \\ \vdots & \vdots \\ \mathbf{u}_m^\top & ({}^0\mathbf{R}_P {}^P\mathbf{b}_m \times \mathbf{u}_m)^\top \end{bmatrix} \quad (2)$$

Where:

$$\mathbf{u}_i = \frac{\mathbf{a}_i - \mathbf{r}_i}{\|\mathbf{a}_i - \mathbf{r}_i\|}$$



Kinematic Modeling

$$\dot{\rho} = \mathbf{J}_a \dot{\mathbf{a}} - \mathbf{J}_p \xi = \mathbf{J}_a \begin{bmatrix} \dot{\mathbf{a}}^{(a)} \\ \dot{\mathbf{a}}^{(g)} \end{bmatrix} - \mathbf{J}_p \xi \quad (1)$$

Static Modeling

$$\begin{aligned} \underbrace{\tau^\top \dot{\rho}}_{\text{cable power}} &= \underbrace{\tau^\top \mathbf{J}_a \dot{\mathbf{a}}}_{\text{actuator power}} - \underbrace{\tau^\top \mathbf{J}_p \xi}_{\text{payload power}} \\ &= (\mathbf{J}_a^\top \tau)^\top \dot{\mathbf{a}} - (\mathbf{J}_p^\top \tau)^\top \xi \\ &= \mathbf{F}_a^\top \dot{\mathbf{a}} - \mathbf{W}_p^\top \xi \end{aligned}$$

Where:

$$\mathbf{F}_a = \mathbf{J}_a^\top \tau, \quad \mathbf{W}_p = \mathbf{J}_p^\top \tau \quad (2)$$

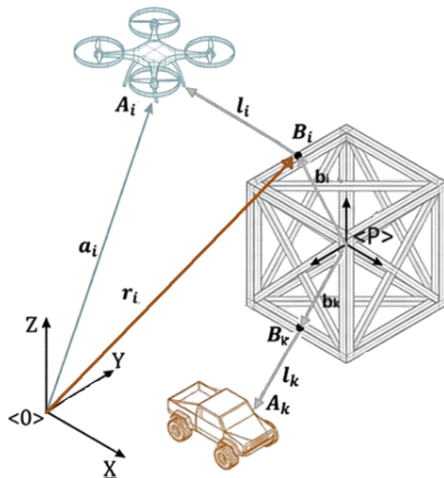
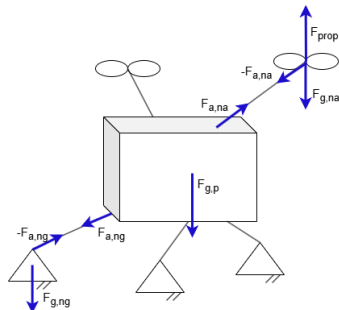


Image source: AI generated

Kinematostatic System Matrix

$$\mathbf{W}_{sys} = \begin{bmatrix} \mathbf{W}_p \\ \mathbf{F}_a \end{bmatrix} = \begin{bmatrix} \mathbf{J}_p^T \\ \mathbf{J}_a \end{bmatrix} \boldsymbol{\tau} = \mathbf{G} \boldsymbol{\tau}$$

$$= \begin{bmatrix} -\mathbf{W}_g - \mathbf{W}_{cnt} - \mathbf{W}_{wind} \\ \mathbf{F}_{prop} + \mathbf{F}_{grd} + \mathbf{F}_g \end{bmatrix}$$


Payload

$$\mathbf{W}_p + \mathbf{W}_g + \mathbf{W}_{contact} + \mathbf{W}_{wind} = \mathbf{0}$$

**At least 6 DoF for
tension equilibrium**
***i*-th UAV**

$$-\mathbf{F}_{a,i} + \mathbf{F}_{prop,i} + \mathbf{F}_{g,i} = \mathbf{0}$$

Thrust limits
***j*-th UGV**

$$-\mathbf{F}_{a,j} + \mathbf{F}_{ground,j} + \mathbf{F}_{g,j} = \mathbf{0}$$

**Friction
constraints**

Design space – number and position of:

- Aerial actuators
 - Ground actuators
 - Payload attachment points
- and the ground cable routing

This work – Gough–Stewart-inspired, 9 actuators:

- 3 UAVs – anti-gravitational force
- 6 ground cables – full payload pose control

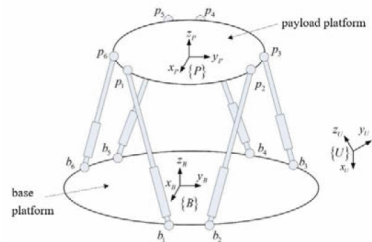
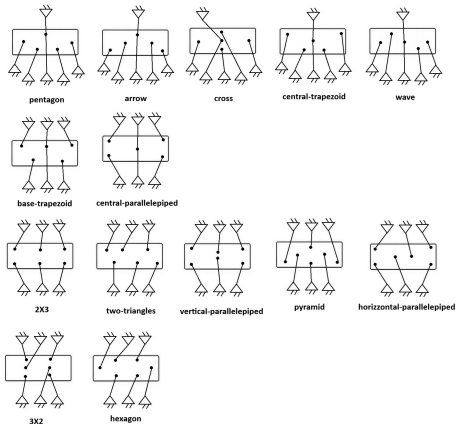
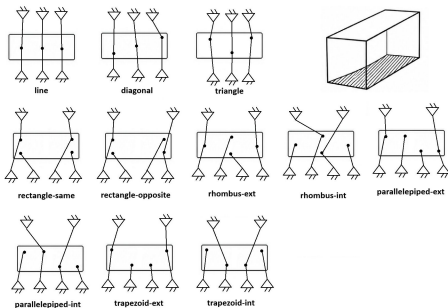
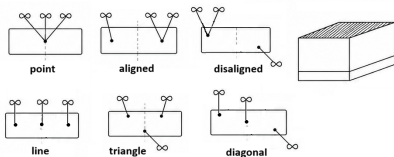


Image source: Wang et al. (2023)

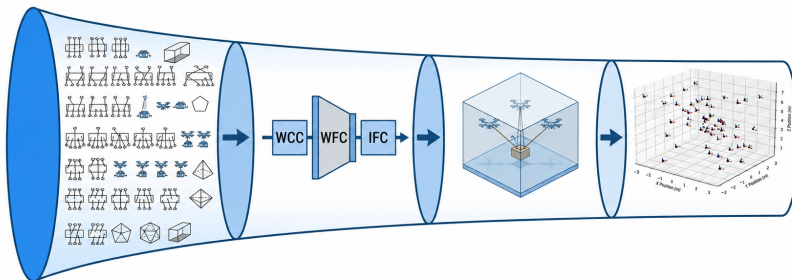
Ground anchors



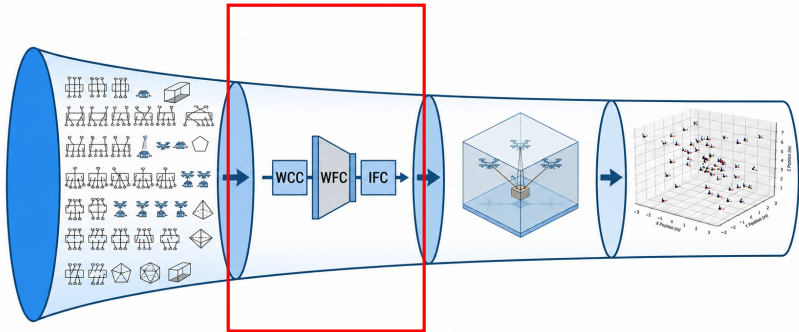
UAV attachment topologies



- Pre-screening (feasibility filters)
- Dynamic simulation (candidate poses)
- Statistical comparison across architectures



- **Pre-screening** (feasibility filters)
- Dynamic simulation (candidate poses)
- Statistical comparison across architectures



Wrench Closure Condition (WCC)

Payload fully constrained: an arbitrary wrench can be balanced with strictly positive tensions.

$$\text{rank}(\mathbf{J}_p) = 6, \quad \exists \boldsymbol{\tau} > \mathbf{0} : \mathbf{J}_p^\top \boldsymbol{\tau} = \mathbf{W}_p^* \quad (3)$$

Wrench Feasible Condition (WFC)

Cable tensions stay within admissible bounds, accounting for UAV thrust and UGV friction limits.

$$\mathbf{0} \leq \boldsymbol{\tau}_{\min} \leq \boldsymbol{\tau} \leq \boldsymbol{\tau}_{\max} \quad (4)$$

Interference-Free Condition (IFC)

No cable-cable rubbing and no collision between cables and payload.

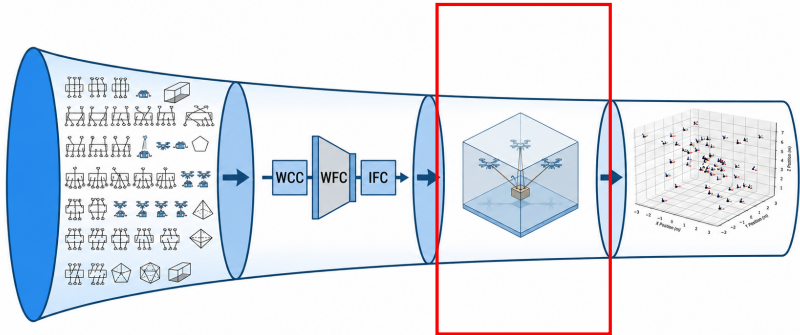
$$d_{ij} \geq d_{\text{safe}}, \quad d_i \geq d_{\text{safe}} \quad \forall i, j \quad (5)$$

Metric		When used
<i>Routing screening</i>		
Conditioning index	$\nu = \frac{\sigma_{\min}(\mathbf{J}_p)}{\sigma_{\max}(\mathbf{J}_p)}$	Rank candidate routings
Manipulability	$w = \sqrt{\det(\mathbf{J}_p \mathbf{J}_p^T)}$	Secondary check on wrench volume
<i>Final architecture evaluation</i>		
Capacity margin	γ	Feasibility w.r.t. task wrench
Worst-case margin	ζ	Distance from actuator saturation
Radius of AWS	r_{AW}	Uniform force/moment capability
<i>Ranking</i>		
Composite score	N	Weighted combination of the above

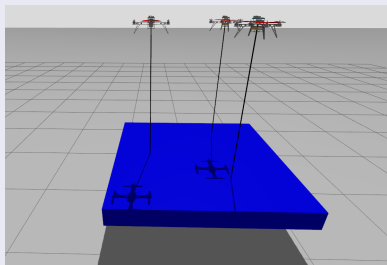
Algorithm 1: Architecture Selection Strategy

```
1: for all payload–ground layout pairs do
2:   for all possible cable routing do
3:     for all test poses do
4:       Build  $J_{p,ground}$ 
5:       Check WCC and IFC ▷ Eq. (3) and (5)
6:       if WCC or IFC fails then
7:         Reject routing and continue
8:       end if
9:     end for
10:    Compute the worst-case conditioning index
11:  end for
12:  Select routing with largest  $\nu$ 
13:  Evaluate WFC using nominal UAV positions ▷ Eq. (4)
14:  if WFC fails then
15:    Optimize UAV positions
16:    Re-evaluate WFC
17:  end if
18:  Accept architecture if WFC is satisfied
19: end for
```

- Pre-screening (feasibility filters)
- **Dynamic simulation** (candidate poses)
- Statistical comparison across architectures

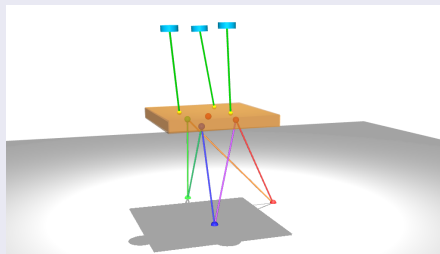


Gazebo Harmonic



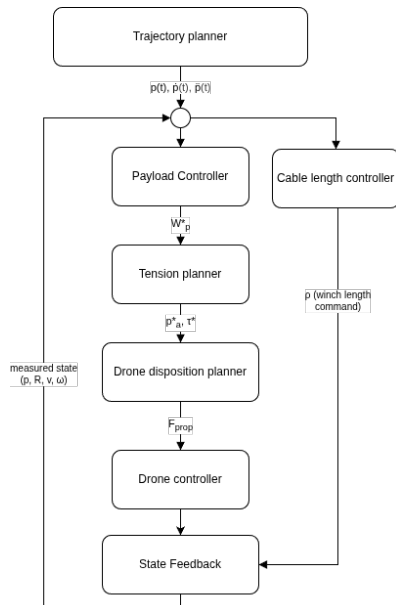
- ROS2 integration, UAV models
- **No native closed-loop** architecture
- Rigid-link cable approximation: slow

MuJoCo



- Native **tendon elements**
- Fast, no rigid-link workaround needed

- **Centralized** architecture
- Two coupled subsystems
 - **Ground winches:** control payload pose relative to the anchors
 - **UAVs:** control to keep the ground cables taut
- Coordination is achieved through:
 - 1 Payload controller
 - 2 Tension planner
 - 3 Winch & drone-level controllers



Payload Controller

$$\mathbf{F}_p^* = m_p \mathbf{g} + \mathbf{K}_p (\mathbf{p}^* - \mathbf{p}) + \mathbf{K}_d (\dot{\mathbf{p}}^* - \dot{\mathbf{p}})$$

$$\mathbf{M}_p^* = \mathbf{R} (-K_R \mathbf{e}_R - K_\Omega \mathbf{e}_\Omega)$$

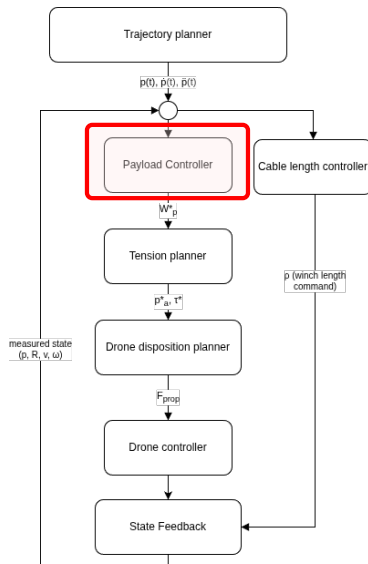
Tension Planner

$$\min_{\mathbf{p}_a} \sum_{i=1}^{n_a} \|\mathbf{F}_{\text{prop},i}(\mathbf{p}_{a,i}, \tau_{\text{optimal},i})\|^2$$

$$+ K \|\mathbf{J}_p^T \boldsymbol{\tau}_{\text{optimal}} - \mathbf{W}_p^*\|^2$$

Winch & Drone-level Controllers

- Ground: $|\Delta \rho_i| \leq v_{\max} \Delta t$
- UAV: $\mathbf{F}_{\text{prop},i} = \mathbf{J}_{a,i}^T \tau_i + m_i g \mathbf{e}_3$

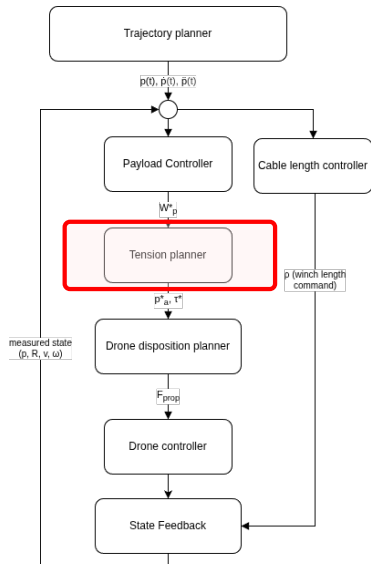


Tension Planner–UAV Placement

$$\begin{aligned}
\min_{\mathbf{p}_a} \quad & \sum_{i=1}^{n_a} \|\mathbf{F}_{\text{prop},i}(\mathbf{p}_{a,i}, \tau_{\text{optimal},i})\|^2 \\
& + K \left\| \mathbf{J}_p^\top \tau_{\text{optimal}} - \mathbf{W}_p^* \right\|^2 \\
\text{s.t.} \quad & x_i^2 + y_i^2 + z_i^2 = l_i^2, \quad \forall i \in [1, n_a] \\
& \rho_{\text{WCC}}(\mathcal{U}) \leq 0, \\
& \rho_{\text{IFC}}(\mathcal{U}) \leq 0.
\end{aligned}$$

Tension Planner – τ Allocation

$$\begin{aligned}
\tau_{\text{optimal}} &= \arg \min_{\tau} \frac{1}{2} \tau^\top \mathbf{H} \tau \\
\text{s.t.} \quad & \mathbf{J}_p^\top(\mathbf{u}) \tau = \mathbf{W}_p^* \\
& \tau_{\min,i} \leq \tau_i \leq \tau_{\max,i}, \forall i \in [1, n_a + n_g].
\end{aligned} \tag{6}$$



Payload Controller

$$\mathbf{F}_p^* = m_p \mathbf{g} + \mathbf{K}_p (\mathbf{p}^* - \mathbf{p}) + \mathbf{K}_d (\dot{\mathbf{p}}^* - \dot{\mathbf{p}})$$

$$\mathbf{M}_p^* = \mathbf{R} (-K_R \mathbf{e}_R - K_\Omega \mathbf{e}_\Omega)$$

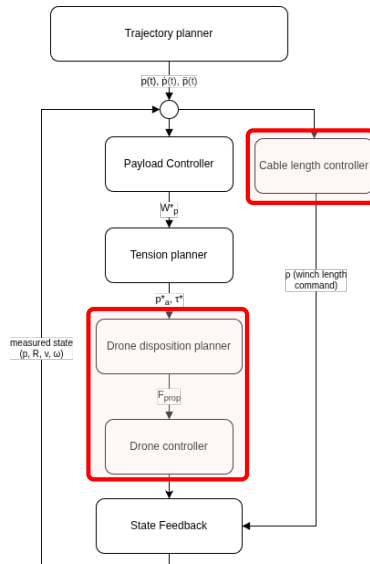
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$$\min_{\mathbf{p}_a} \sum_{i=1}^{n_a} \|\mathbf{F}_{\text{prop},i}(\mathbf{p}_{a,i}, \tau_{\text{optimal},i})\|^2$$

$$+ K \|\mathbf{J}_p^T \boldsymbol{\tau}_{\text{optimal}} - \mathbf{W}_p^*\|^2$$

Winch & Drone-level Controllers

- Ground: $|\Delta \rho_i| \leq v_{\max} \Delta t$
- UAV: $\mathbf{F}_{\text{prop},i} = \mathbf{J}_{a,i}^T \tau_i + m_i \mathbf{g} \mathbf{e}_3$



Minimal tension ($\tau_i = \tau_{\min} \forall i$)

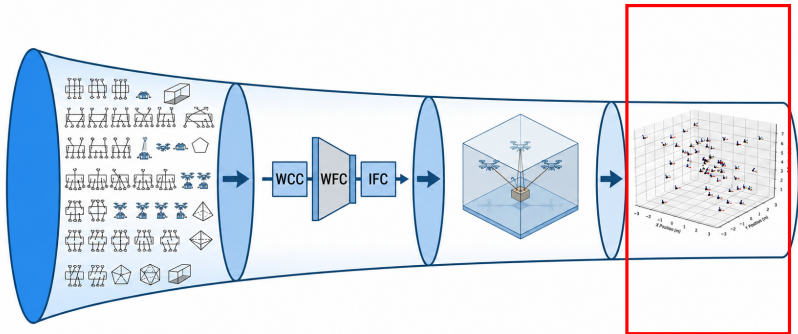
- No freedom to redistribute tensions
- AWS collapses to a single point

Optimal tension (Eq. (6))

- Tensions vary within $[\tau_{\min}, \tau_{\max}]$
- Larger capacity margin
- Adopted for all subsequent simulations

Metric	$\tau_{\min}$	τ_{opt}
Conditioning index ν	0.102	0.096
Manipulability w	0.477	0.437
Radius $r_{AW,force}$	49.297	47.903
Capacity margin γ	-0.193	10.464
Worst-case capacity margin ζ	0.909	0.764
Composite score N	17.351	20.628
Position Error (m)	0.002	0.003
Orientation Error (rad)	0.001	0.002

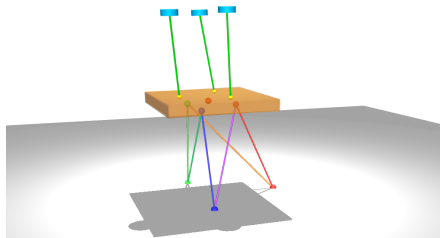
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- Dynamic simulation (candidate poses)
- **Statistical comparison across architectures**



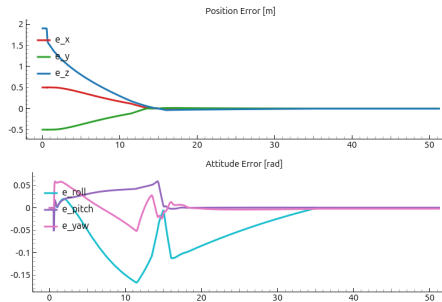
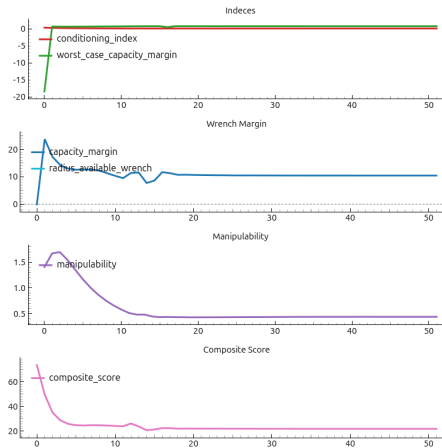
Why Stewart?

- Symmetric architecture
- Well-studied in the literature
- Reference point for alternative ACTS architectures

Numerically verified to confirm implementation correctness before comparing architectures



Metric	Value
Conditioning ν	0.096
Manipulability w	0.437
Capacity margin γ	10.46
Position error	0.003 m
Orientation error	0.002 rad



Simulation parameters

Parameter	Value
Payload mass	1 kg
Min / max cable tension	5.5 N / 150 N
Max winch speed	1.5 m/s
Max rotor velocity	1666 rad/s
Min UAV–UAV distance	0.4 m
Min cable–cable distance	0.05 m
Position tolerance	0.1 m
Orientation tolerance	3°

Statistical methodology

- Same 60 poses for every architecture → paired tests
- Significance level $\alpha = 0.05$
- **Omnibus:** Friedman (continuous) / Cochran's Q (binary)
- **Post-hoc** vs. `acts_stewart`: Wilcoxon / McNemar
- **Correction:** Benjamini–Hochberg (omnibus), Holm (post-hoc)

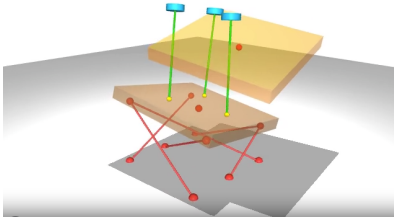
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- Ground Anchor Spacing – wider footprint, better performance
- UAV Attachment Geometry – the dominant architectural factor

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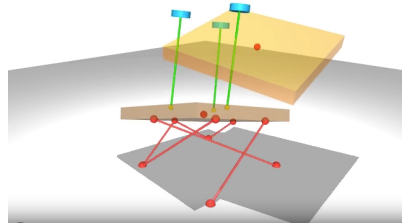
- Static indices are **necessary but not sufficient** for closed-loop performance
- Best static scores $\neq$ best pose reachability
- Worst static scores $\neq$ worst dynamic behavior

Architecture	ν	w	N	Pose Reach.	Cable Rub
rectangle-opposite-2x3	0.1246 (1)	1.0459 (1)	8.9142 (1)	0.5833 (6)	0.9333 (5)
two-triangles-rect-same	0.1052 (3)	0.4603 (3)	7.3013 (2)	0.6500 (4)	1.0000 (6)
parallel-int-2x3	0.0997 (4)	0.4536 (4)	6.7857 (3)	0.7500 (2)	0.4667 (2)
pentagon-rect-same	0.0869 (5)	0.3522 (5)	6.4366 (4)	0.8333 (1)	0.6500 (3)
two-triangles-2x3	0.1140 (2)	0.5953 (2)	6.2580 (5)	0.6667 (3)	0.7667 (4)
acts_stewart	0.0533 (6)	0.1054 (6)	5.5706 (6)	0.6500 (4)	0.3167 (1)

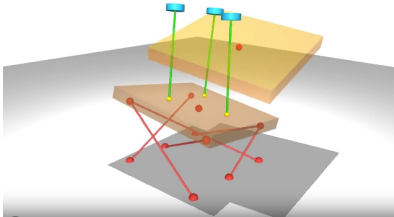
Best static, worst dynamic



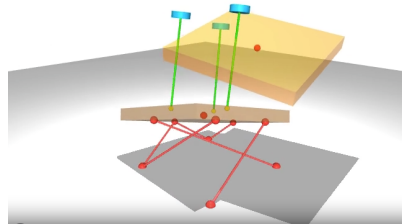
4th-ranked static, best dynamic



Best static, worst dynamic



4th-ranked static, best dynamic



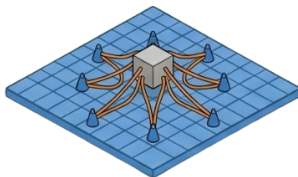
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- **Ground Anchor Spacing** – wider footprint, better performance
- UAV Attachment Geometry – the dominant architectural factor

Pros: Wider spacing $\rightarrow$ larger moment arms $\rightarrow$ wider achievable wrench set

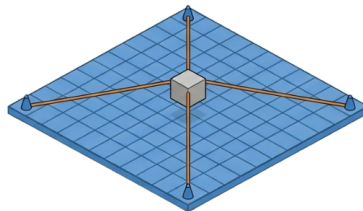
Trade-off with on-site space

Metric	Small	Nominal	Large
Capacity margin γ	0.5533	2.7714	6.7333
Conditioning index ν	0.0398	0.0533	0.0788
Manipulability w	0.0551	0.1054	0.2872
Pose reached	46.7%	65.0%	88.3%
Cable rubbed	53.3%	31.7%	20.0%
Unreached poses	10/60	5/60	0/60

Small footprint



Large footprint

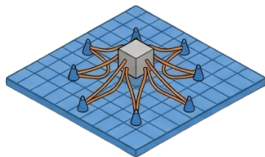


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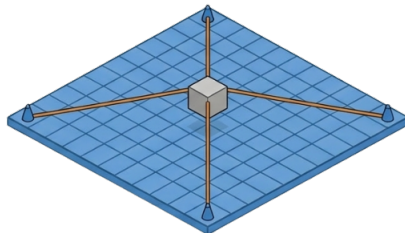
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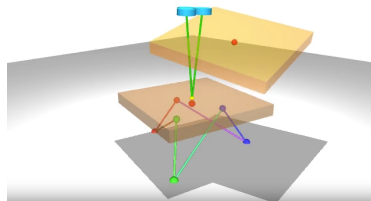
Large footprint



- Static vs. Dynamic – static indices alone don't predict task success
- Ground Anchor Spacing – wider footprint, better performance
- **UAV Attachment Geometry** – the dominant architectural factor

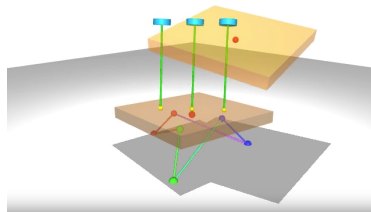
- **0D attachment:** UAVs lose independent moment generation about the payload
- **1D attachment:** moments restored along one axis only – still limited reachability
- **2D attachment:** independent moment arms in two directions → full six-DoF wrench authority

Point Configuration



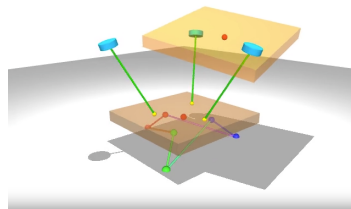
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Diagonal Configuration



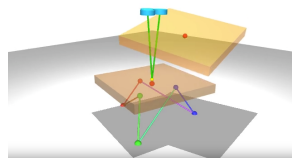
- **0D attachment:** UAVs lose independent moment generation about the payload
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Triangle Configuration

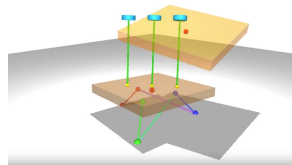


Configuration		Pose Reach.	Cable Rubbing
Point		0%	95.3%
Line		4.0%	88.3%
Diagonal		21.7%	65.0%
Aligned		9.0%	76.7%
Disaligned		16.7%	68.3%
Triangle (Stewart)		65.0%	31.7%

Point Configuration



Diagonal Configuration



- Kinemastostatic model developed for hybrid ACTS
- Centralized controller implemented and validated
- Selection methodology combining feasibility filters (WCC, WFC, IFC) with geometric performance indices
- Validated across 60 test poses in MuJoCo

Key findings

- **Static indices:** necessary but not sufficient for closed-loop performance
- **Ground-anchor spacing:** monotonic benefit
- **UAV attachment geometry:** triangular topology outperforms other layouts

- Sample candidate poses along the actual desired trajectory rather than randomly
- Explore different numbers of UAVs / UGVs and multi-cable UAV connections
- Consider attachment points on arbitrary payload faces
- Model physical interaction with a vertical surface (façade contact)
- Validate experimentally on a real ACTS platform: sensing noise, unmodeled aerodynamics, cable elasticity

Thank you.

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